

Backtracking, Branch & Bound

Module 2

Sem IV

AoA Even 2021-22

Introduction

- Backtracking and Branch and Bound are two graph based methods for design of algorithms.
- In both cases we explore a search tree.
- In Backtracking, we start with a node and explore the nodes in **Depth First manner**.
- All the nodes need not to be explored. Cut the branches of the tree based on the constraint of the problem. This reduces the time complexity of the algorithm.
- Branch and Bound explores the search tree in a **Breadth First** manner.

Backtracking : Introduction

- It is modified form of **Depth First Search**.
- Here solution vector is of form $x_1, x_2, x_3, \dots, x_n$, n tuple $(x_1, x_2, x_3, \dots, x_n)$, where x_i is chosen from finite set of S_i , such that constraint of the problem is satisfied.
- Backtracking algorithm solves the problem using two types of constraints:
 1. Explicit Constraint
 2. Implicit Constraint

Backtracking : Introduction

Terminologies Used:

Backtracking algorithms determine problem solutions by systematically searching for solution using tree structure.

1. **State space tree:** The solution space is organized as a tree called the state space tree.
2. **Explicit constraint:** These are the rules that restrict each component x_i of the solution vector to take values only from a given set S .
3. **Implicit constraint:** These are the rules that describe the way in which the x_i 's must relate to each other or which of the components of the solution vector satisfy the criteria function.
4. **Solution space:** It is the set of all tuples that satisfy the explicit constraints.
5. **Live node:** It is the node that has been generated, but none of its descendants are yet generated.
6. **Bounding function or criteria:** It is a function created that is used to kill live nodes without generating all its children.

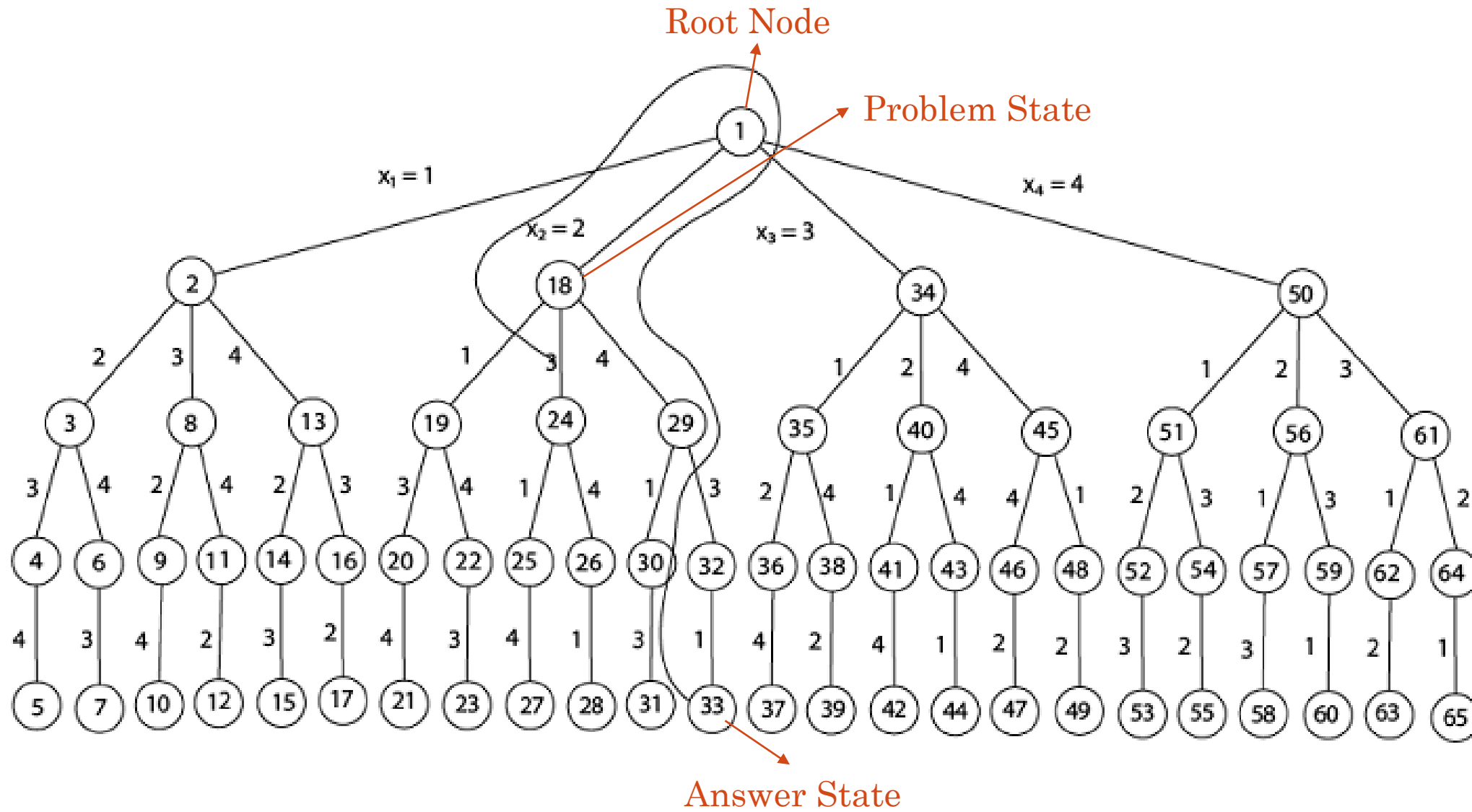
Backtracking : Introduction

Terminologies Used:

Backtracking algorithms determine problem solutions by systematically searching for solution using tree structure.

7. **Extended node or E-node:** It is the live node whose children are currently being generated.
8. **Dead node:** It is the node that is not to be extended further or all of whose children have already been generated.
9. **Answer node:** It is the node that represents the answer of the problem that means the node at which the criteria functions are maximized, minimized or satisfied.
10. **Solution node:** It is the node that has the possibility to become the answer node.

Backtracking : Introduction



Backtracking : N-Queen Problem

1. 2-Queen problem
2. 3-Queen Problem
3. 4-Queen Problem
4. 8-Queen problem

2-Queen problem:

Q	x
x	x

x	Q
x	x

Therefore, No Solution

Backtracking : N-Queen Problem

1. 2-Queen problem
2. 3-Queen Problem
3. 4-Queen Problem
4. 8-Queen problem

2-Queen problem:

Q	x
x	x

x	Q
x	x

Therefore, No Solution

Similarly, No Solution for 3-Queen problem

Backtracking : N-Queen Problem

4 Queen Problem:

State space Tree:

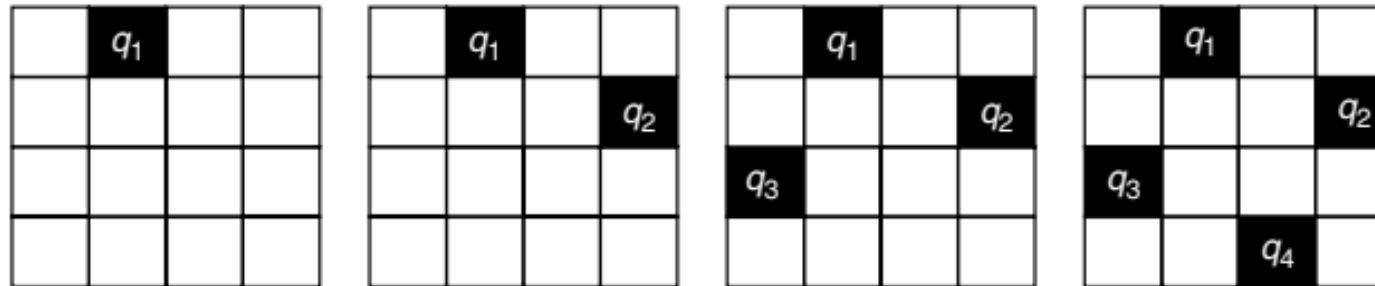


Figure 2 Solution of four-queens problem.

Backtracking : N-Queen Problem

4 Queen Problem:

Solution can be represented as four-tuple (x_1, x_2, x_3, x_4) where x_1 is column value in row 1 for placement of Q_1 and so on.

	q_1		
			q_2
q_3			
		q_4	

Solution 2, 4, 1, 3

		q_1	
q_2			
			q_3
	q_4		

Solution 3, 1, 4, 2

Figure 7 Two possible solutions to four-queens problem.

Backtracking : N-Queen Problem

N Queen Problem:

```
1  Algorithm NQueens( $k, n$ )
2  // Using backtracking, this procedure prints all
3  // possible placements of  $n$  queens on an  $n \times n$ 
4  // chessboard so that they are nonattacking.
5  {
6      for  $i := 1$  to  $n$  do
7      {
8          if Place( $k, i$ ) then
9          {
10              $x[k] := i$ ;
11             if ( $k = n$ ) then write ( $x[1 : n]$ );
12             else NQueens( $k + 1, n$ );
13         }
14     }
15 }
```

Algorithm 7.5 All solutions to the n -queens problem

Backtracking : N-Queen Problem

N Queen Problem:

```
1  Algorithm Place( $k, i$ )
2  // Returns true if a queen can be placed in  $k$ th row and
3  //  $i$ th column. Otherwise it returns false.  $x[ ]$  is a
4  // global array whose first  $(k - 1)$  values have been set.
5  // Abs( $r$ ) returns the absolute value of  $r$ .
6  {
7      for  $j := 1$  to  $k - 1$  do
8          if  $((x[j] = i) // \text{Two in the same column}$ 
9              or  $(\text{Abs}(x[j] - i) = \text{Abs}(j - k)))$ 
10             // or in the same diagonal
11             then return false;
12      return true;
13 }
```

Algorithm 7.4 Can a new queen be placed?

Backtracking : N-Queen Problem

N Queen Problem:

- (4,1), (5,2), (6,3), (7,4), (8,5)

Let (i, j) and (k, l) be two cells in the chessboard.

If $i - j = k - l$

e.g. $4 - 1 = 6 - 3 = 13$

Rearranging above equation, we have

$$i - k = j - l \Rightarrow |i - k| = |j - l|$$

- (5,8), (6,7), (7,6), (8,5)

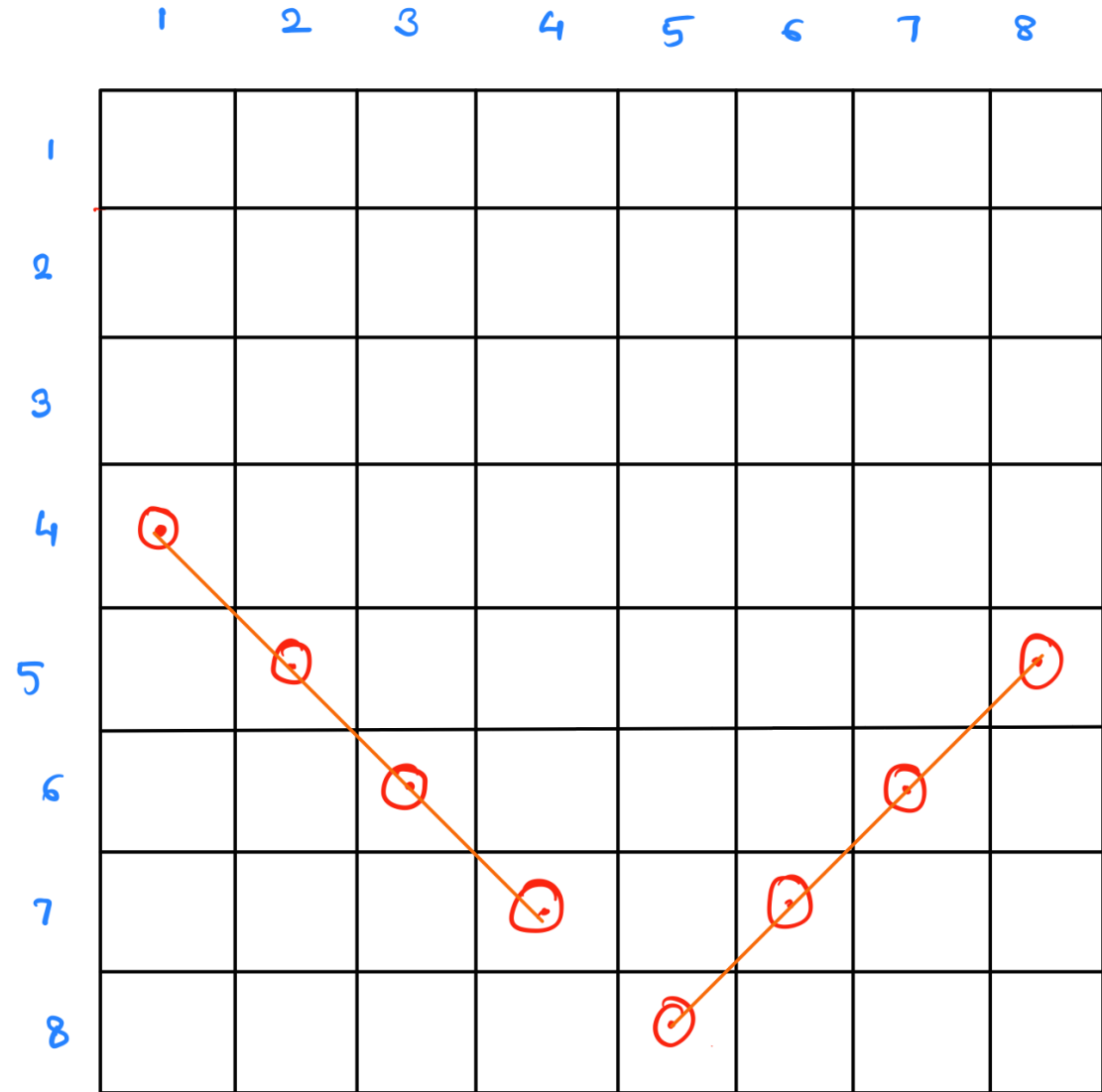
Again let (i, j) and (k, l) be two cells in the chessboard.

If $i + j = k + l$

e.g. $5 + 8 = 6 + 7 = 13$

Rearranging above equation, we have

$$i - k = l - j \Rightarrow |i - k| = |j - l|$$



Backtracking : Sum of Subsets

Given: 1. n distinct positive numbers (called weights w_i), where $1 \leq i \leq n$
2. Sum (m)

We need to find all possible subsets of given numbers (w_i) having sum equal to the target Sum (m).

Backtracking : Sum of Subsets

Example:

$n=4; (w_1, w_2, w_3, w_4) = (11, 13, 24, 7) \text{ \& } m=31$

Desired subsets are (11,13,7) & (24,7)

Solution vector (1, 2, 4) & (3, 4) $\rightarrow$ [Variable Length]

In general, all solution vectors are k-tuples, $(x_1, x_2, x_3, x_4) ; 1 \leq k \leq n$

Implicit Constraints:

1. No two subsets should be same & sum of corresponding w_i 's be m
2. $x_i < x_{i+1}$ such that $1 \leq i \leq k$, to avoid generating multiple instances of same subset e.g. (1,2,4) & (1,4,2)

Backtracking : Sum of Subsets

Example:

$n=4; (w_1, w_2, w_3, w_4) = (11, 13, 24, 7) \text{ \& } m=31$

Another Approach: **[Fixed Length]**

Each solution set is represented by n-tuple (x_1, x_2, x_3, x_4) such that $x_i \in \{0,1\}$ where $1 \leq i \leq n$

$x_i = 0 \rightarrow w_i \text{ not selected, and } x_i = 1 \rightarrow w_i \text{ selected}$

Therefore, Solution space of above instance are $(1,1,0,1) \text{ \& } (0,0,1,1)$

Backtracking : Sum of Subsets

Example:

$n=4; (w_1, w_2, w_3, w_4) = (11, 13, 24, 7) \text{ \& } m=31$

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Backtracking : Sum of Subsets

Example:

$$S = \{2, 3, 5, 6, 7, 9, 10\} \text{ \& } M = 15$$

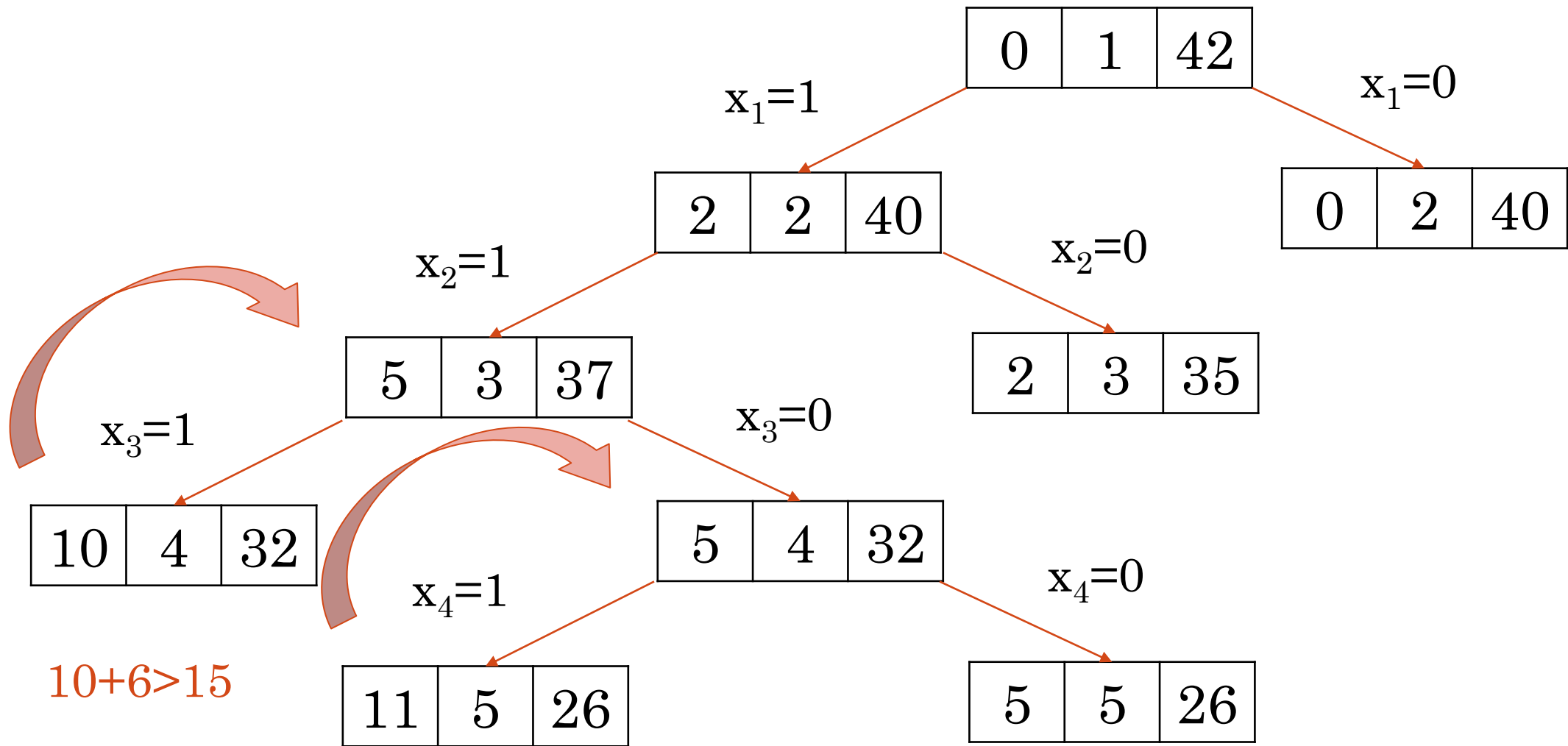
In state space tree of solution, node list values of
sumSoFar, k & remWeight

Initialize root node with values, sumSoFar = 0, k = 1 & remWeight = 42

0	1	42
---	---	----

Backtracking : Sum of Subsets

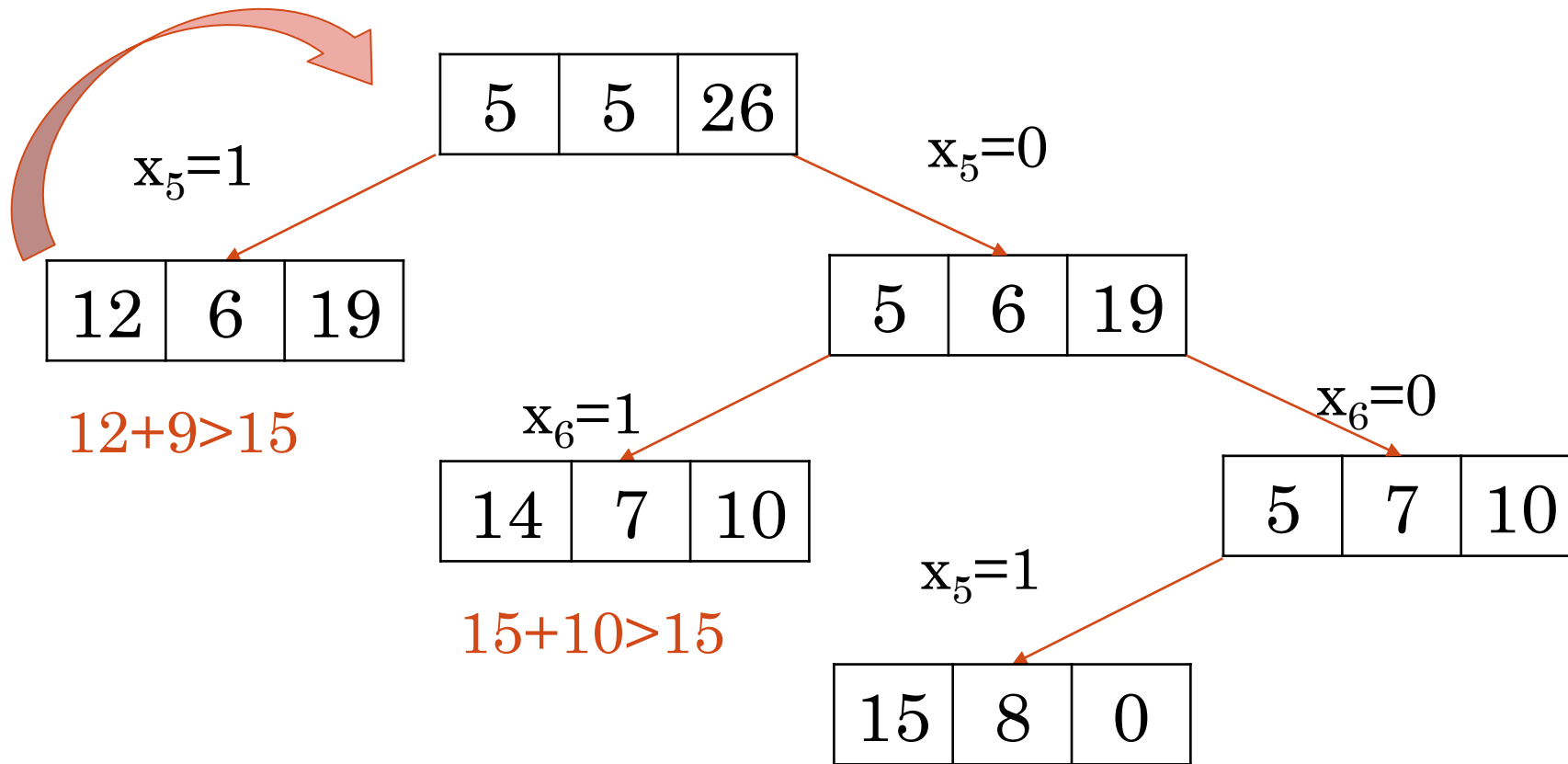
Example:



$S = \{2, 3, 5, 6, 7, 9, 10\}$ & $M = 15$

$11+7 > 15$

Backtracking : Sum of Subsets



$S = \{2, 3, 5, 6, 7, 9, 10\}$ & $M = 15$

Backtracking : Sum of Subset

Algorithm 4 SUMOFSUBSETS (Sumsofar, k, remweight)

This algorithm is used to find all the solutions of the sum of subsets problem. The $X[]$ is the solution vector.

```
1. Set  $X[k]=1$ 
2. if ( $\text{Sumsofar}+w[k]=M$ ) then
    print  $X[1..k]$ 

    // solution is found
    else
        if ( $\text{Sumsofar}+w[k]+w[k+1]\leq M$ ) then
            // Generate Left child
            SUMOFSUBSETS ( $\text{Sumsofar}+w[k]$ ,  $k+1$ ,  $\text{remweight}-w[k]$ )
        Endif
3. Endif
4. if ( $(\text{Sumsofar}+\text{remweight}-w[k]\geq M)$  and  $(\text{Sumsofar}+w[k+1]\geq M)$ ) then
    // Generate Right child
    {
         $X[k]=0$ 
        SUMOFSUBSETS ( $\text{Sumsofar}$ ,  $k+1$ ,  $\text{remweight}-w[k]$ )
    }
5. Stop
```

Backtracking : Graph Coloring

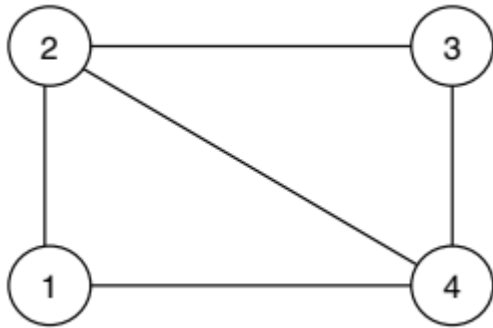
- It's a classic combinatorial Problem
- It's a problem of coloring N vertices of a given graph G in such a way that no two adjacent vertices share the same color and yet M colors are used.
- The problem is called as M coloring problem.
- M coloring Decision problem: M is given, whether graph can be colored using M colors
- M coloring optimization problem: smallest number of colors (M) required to color the graph.

Backtracking : Graph Coloring Algorithm

- Suppose we have graph $G=(V,E)$ with N vertices and M is given number of colors.
- We represent Graph G by adjacency matrix $G[n,n]$ where,
 - $G[i,j]=1$ if (i,j) is an edge of G and
 - $G[i,j]=0$ otherwise.
- If d is degree of given graph, then it can be colored with $d+1$ colors [m is referred to as **chromatic number**].
- Here colors are represented as integers $1,2,3,\dots,M$ and coloring solution will be a vector $x[1\dots N]$.
- So, solutions are given by n -tuple $(x_1, x_2, x_3, \dots, x_n)$ where, $1 \leq x_i \leq M$ and $1 \leq i \leq N$ and x_i is color of node i .

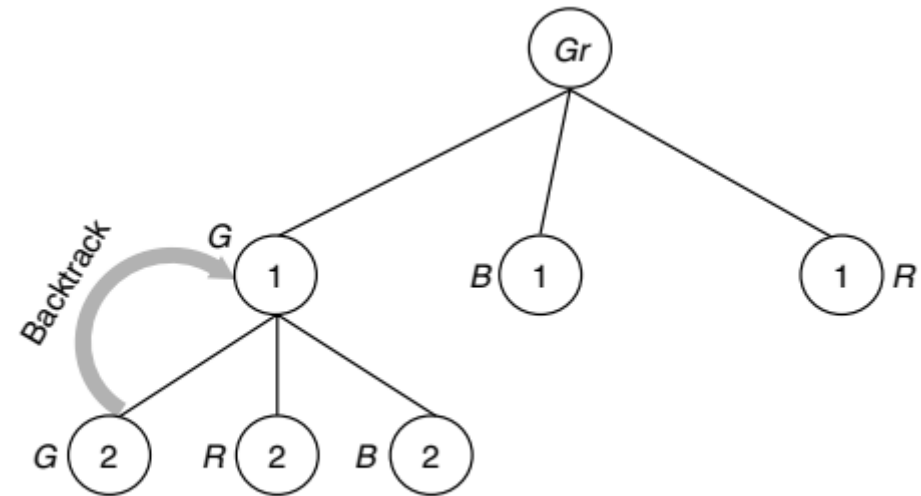
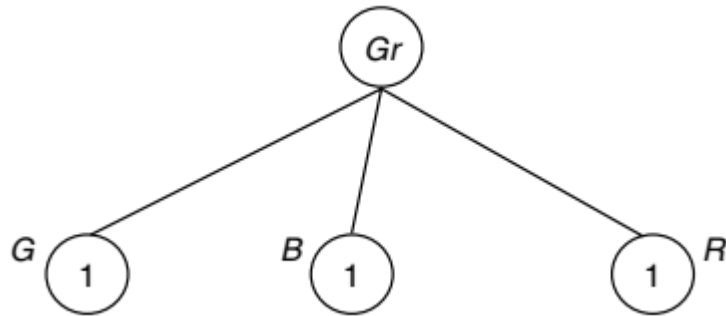
Backtracking : Graph Coloring Example

Example 1:



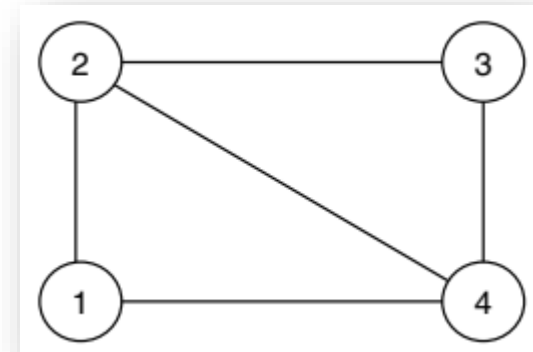
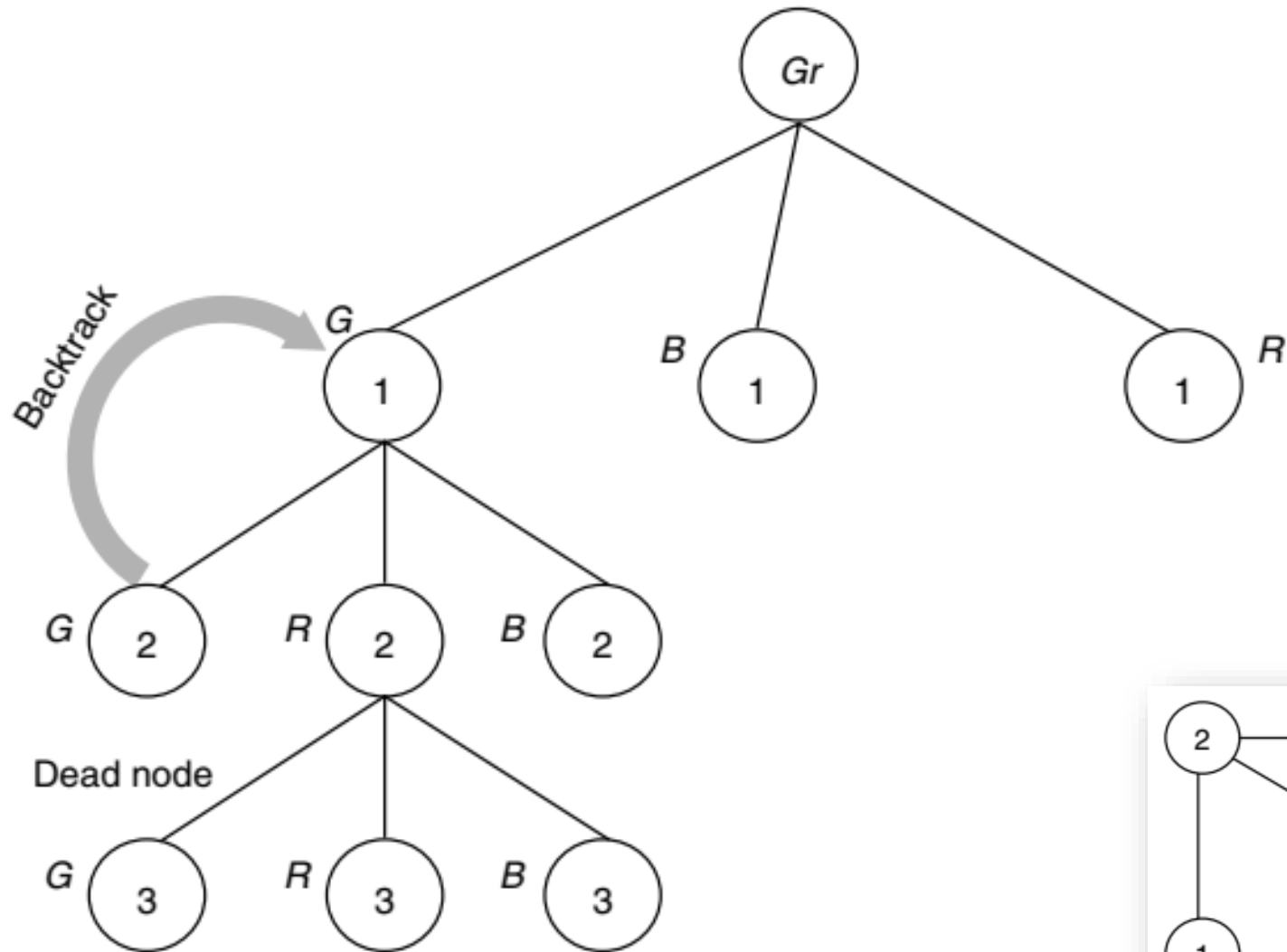
	1	2	3	4
1	0	1	0	1
2	1	0	1	1
3	0	1	0	1
4	1	1	1	0

State Space Tree:

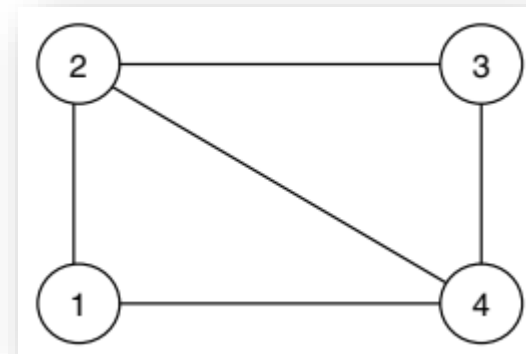
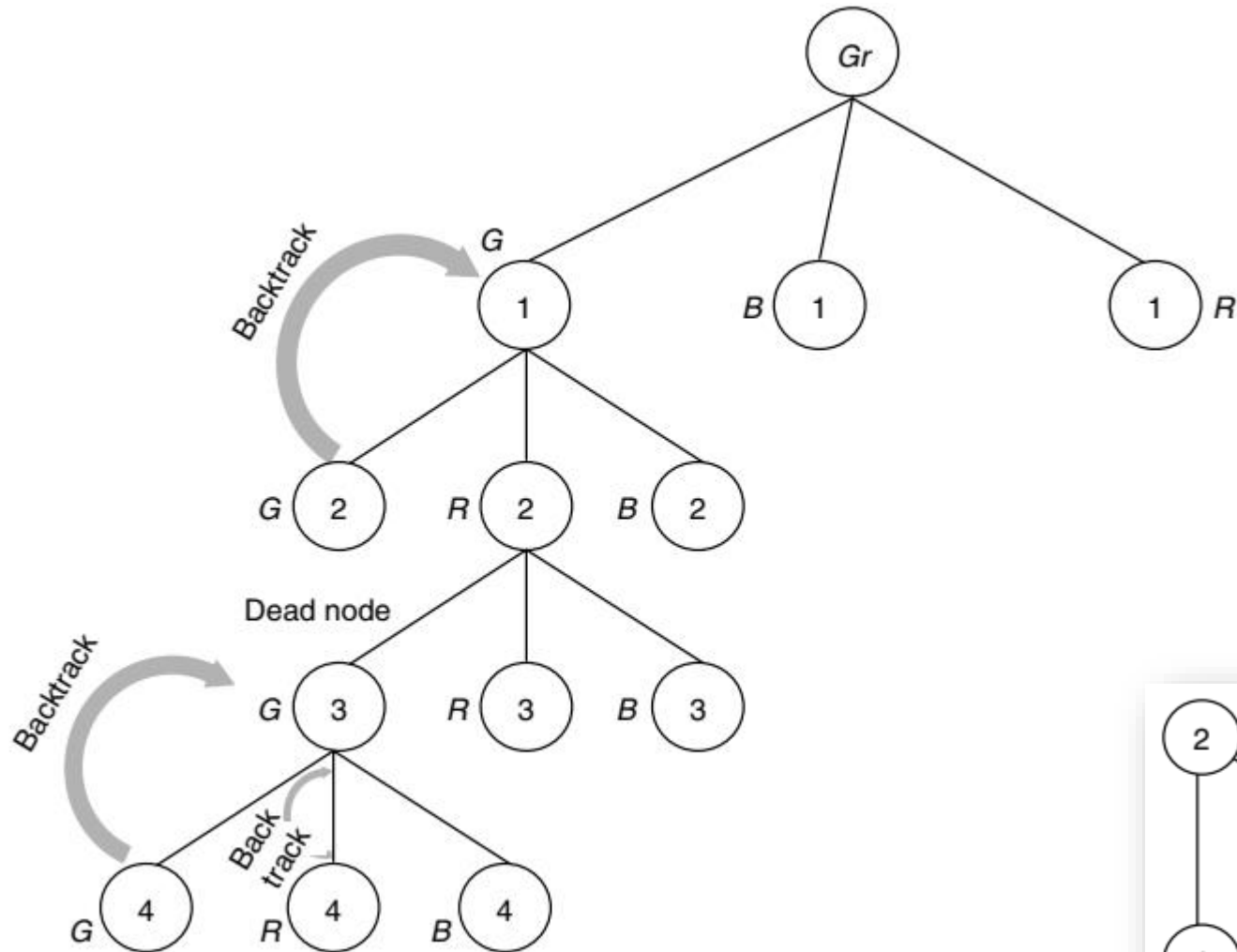


Dead node

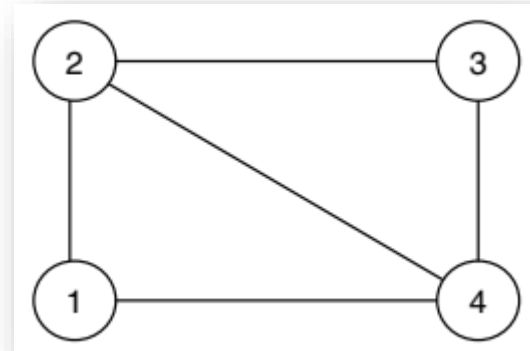
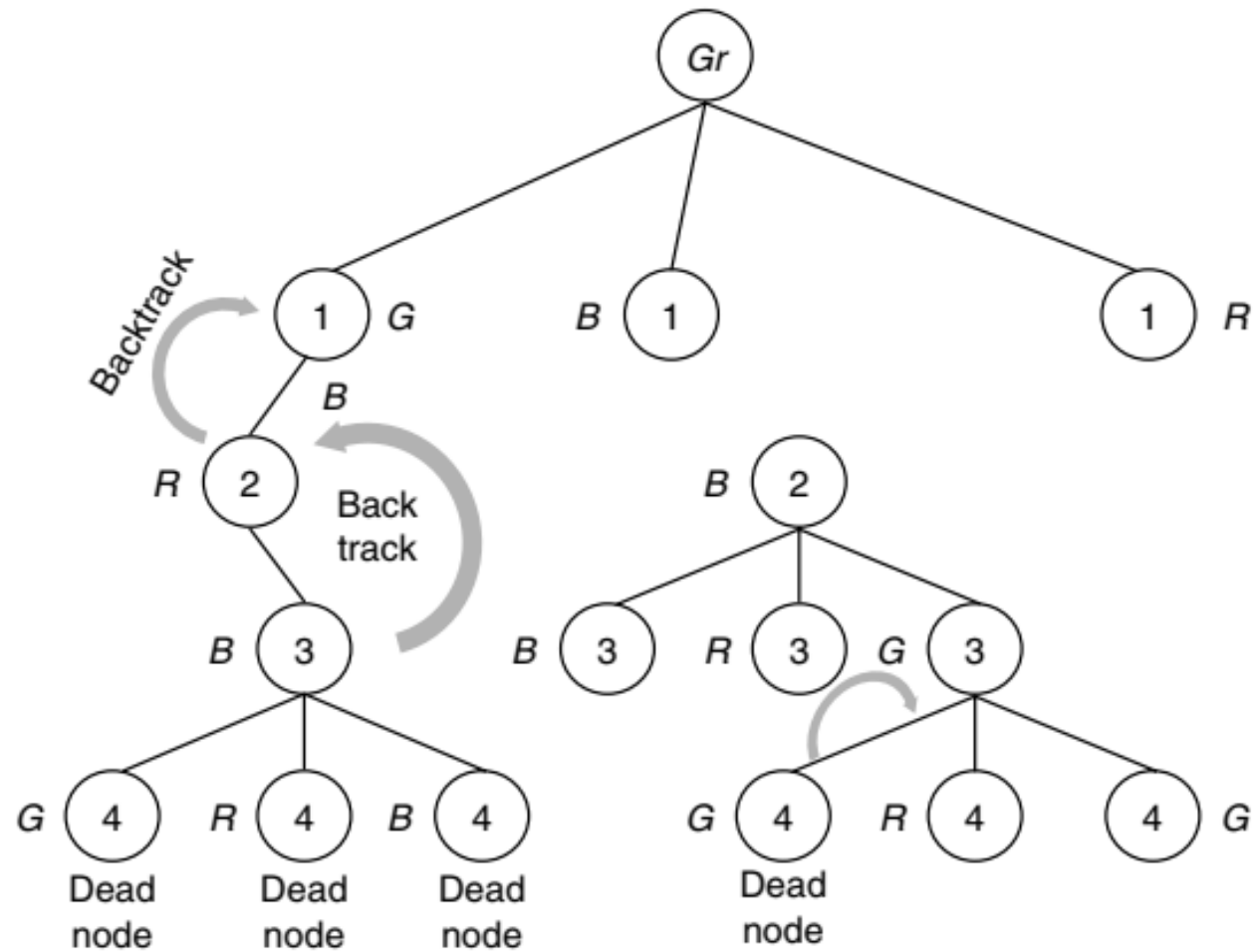
Backtracking : Graph Coloring Example



Backtracking : Graph Coloring Example

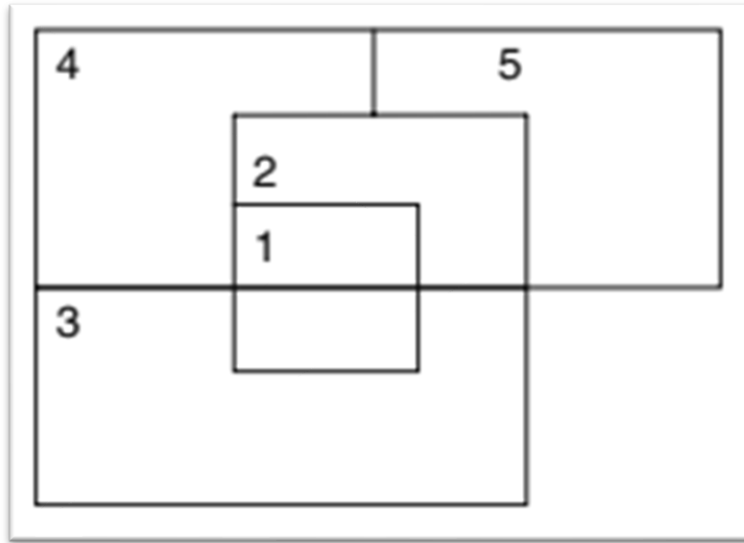


Backtracking : Graph Coloring Example

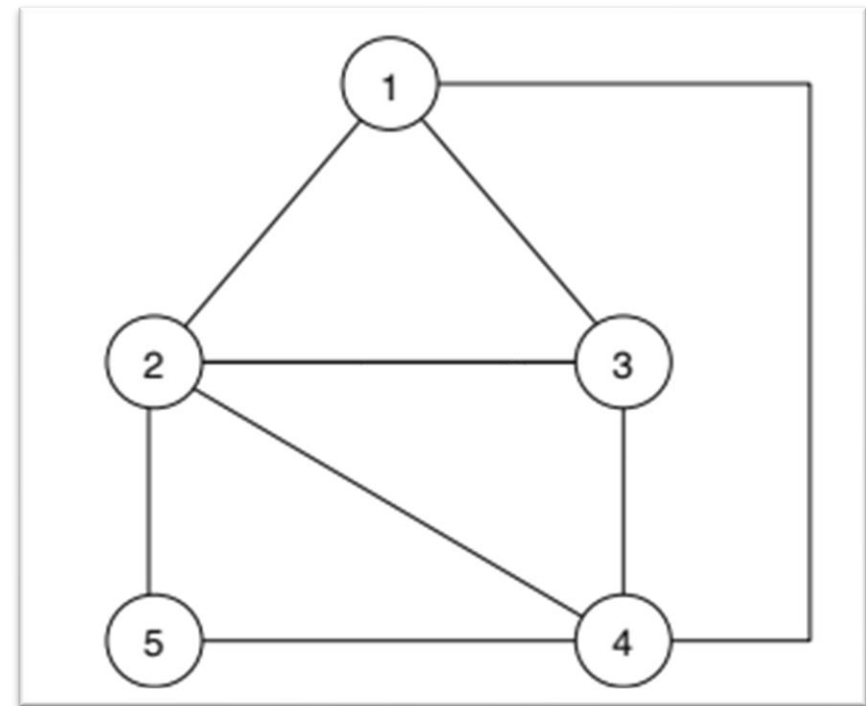
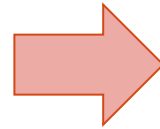


Backtracking : Graph Coloring Example

Example 2:



Map



Planer Graph

Backtracking : Graph Coloring Algorithm

Algorithm mColoring(k)

//The graph is represented by its
//boolean adjacency matrix $G[1:n, 1:n]$.

```
{
  repeat
  {
    NextValue(k);
    if( $x[k]=0$ ) then return;
    if( $k=n$ ) then
      write( $x[1:n]$ );
    else mColoring(k+1);
  }until false;
}
```

Algorithm NextValue(k)

```
{
  repeat
  {
     $x[k] = (x[k]+1) \bmod (m+1)$ ;
    if( $x[k]=0$ ) then return;
    for j = 1 to n do
    {
      if(( $G[k,j] \neq 0$ ) and ( $x[k] = x[j]$ )) then
        break;
    }
    if( $j=n+1$ ) then return;
  }until(false);
}
```

Branch & Bound : Introduction

- **Branch:** using State Space Tree (Similar to Backtracking)
- **Bound:** using Upper and Lower bounds
- Branch and Bound differs from backtracking in the sense that all the children of the E-Node are generated before any other live node becomes the E-Node.
- Branch and Bound is the generalization of both graph search strategies, BFS and DFS.
- The state space tree of the branch and bound method can be constructed using following three strategies:
 - FIFO (First In First Out) search (Implemented using QUEUE)
 - LIFO (Last In First Out) search (Implemented using STACK)
 - LC (Least Cost) search (Implemented using PRIORITY QUEUE)

Branch & Bound : Introduction

FIFO (First In First Out) Branch and Bound

- In FIFO search, queue data structure is used.
- Initially node 1 is taken as the E-node.
- The child nodes of node 1 are generated. All these live nodes are placed in a queue.
- Next the first element in the queue is deleted, i.e. node 2, the child nodes of node 2 are generated and placed in the queue.
- This continues until the answer node is found.

Branch & Bound : FIFO

LIFO (Last In First Out) Branch and Bound

Example: Job sequencing with deadlines problem

- Jobs = {J1, J2, J3, J4}; P = {10, 5, 8, 3}; d = {1, 2, 1, 2}
- Node 1 is the E-node. Child nodes of node 1 are generated and placed in the queue.

2	3	4	5	
---	---	---	---	--

- First element in the queue is deleted, ie., 2 is deleted and its child nodes are generated.

3	4	5	6	7	8	
---	---	---	---	---	---	--

- Similarly, the next element is deleted, ie., 3 and its child nodes are generated and placed in the queue. This is continued until an answer node is reached.

Branch & Bound : FIFO

FIFO (First In First Out) Branch and Bound

Example: Job sequencing with deadlines problem

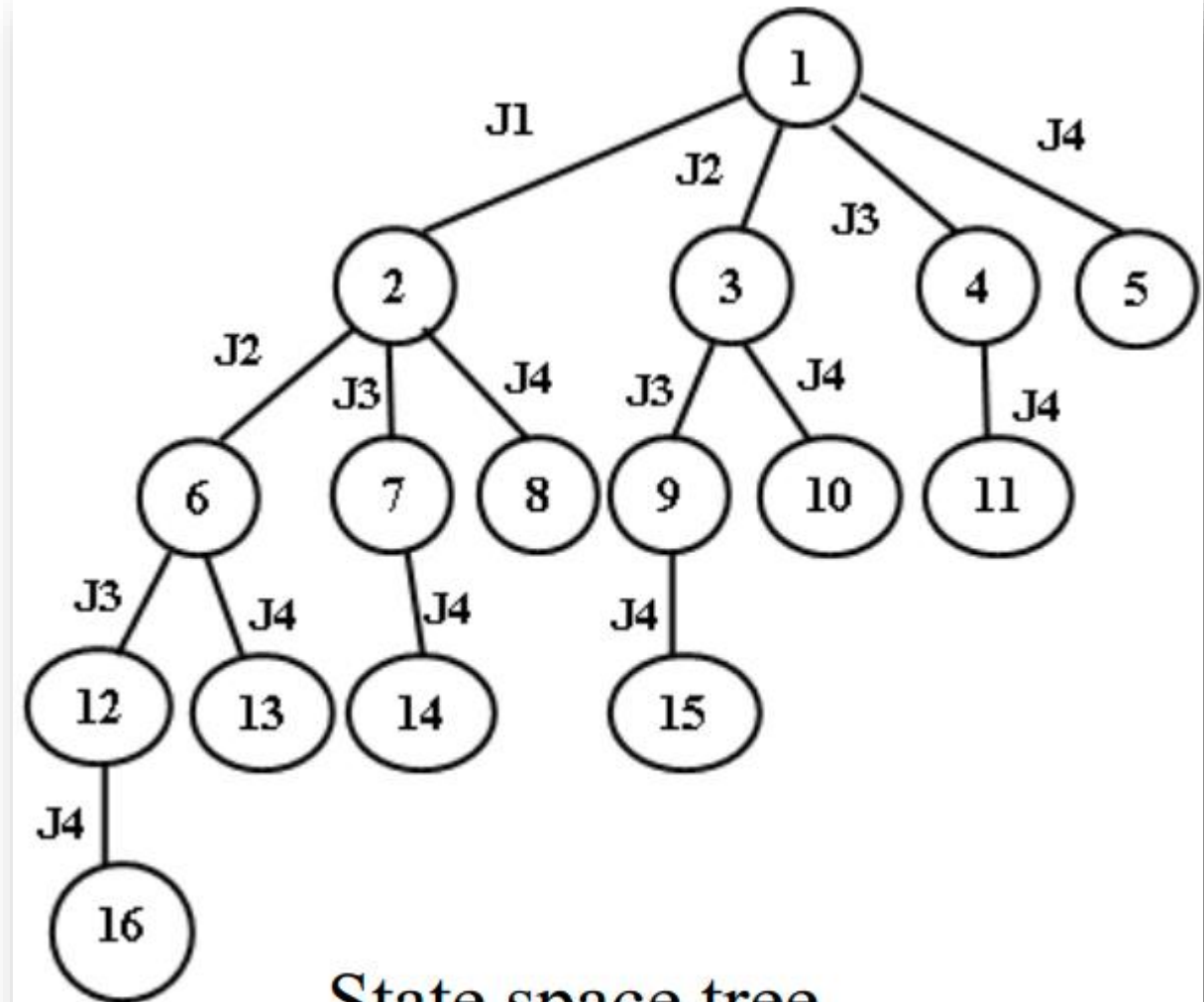
- Jobs = {J1, J2, J3, J4}; P = {10, 5, 8, 3}; d = {1, 2, 1, 2}
- Node 1 is the E-node. Child nodes of node 1 are generated and placed in the queue.

2	3	4	5	
---	---	---	---	--

- First element in the queue is deleted, ie., 2 is deleted and its child nodes are generated.

3	4	5	6	7	8	
---	---	---	---	---	---	--

- Similarly, the next element is deleted, ie., 3 and its child nodes are generated and placed in the queue. This is continued until an answer node is reached.



State space tree

Branch & Bound : LIFO

LIFO (Last In First Out) Branch and Bound

- In LIFO search, stack data structure is used.
- Initially node 1 is taken as the E-node.
- The child nodes of node 1 are generated. All these live nodes are placed in a stack.
- Next the first element in the stack is deleted, i.e. node 5, the child nodes of node 5 are generated and placed in the stack.
- This continues until the answer node is found.

Branch & Bound : LIFO

FIFO (First In First Out) Branch and Bound

Example: Job sequencing with deadlines problem

- Jobs = {J1, J2, J3, J4}; P = {10, 5, 8, 3}; d = {1, 2, 1, 2}
- Node 1 is the E-node. Child nodes of node 1 are generated and placed in the stack.
- First element in the stack is deleted, i.e., 5 is deleted and its child nodes are generated.
- Similarly, the next element is deleted, i.e., 4 and its child nodes are generated and placed in the queue. This is continued until an answer node is reached.

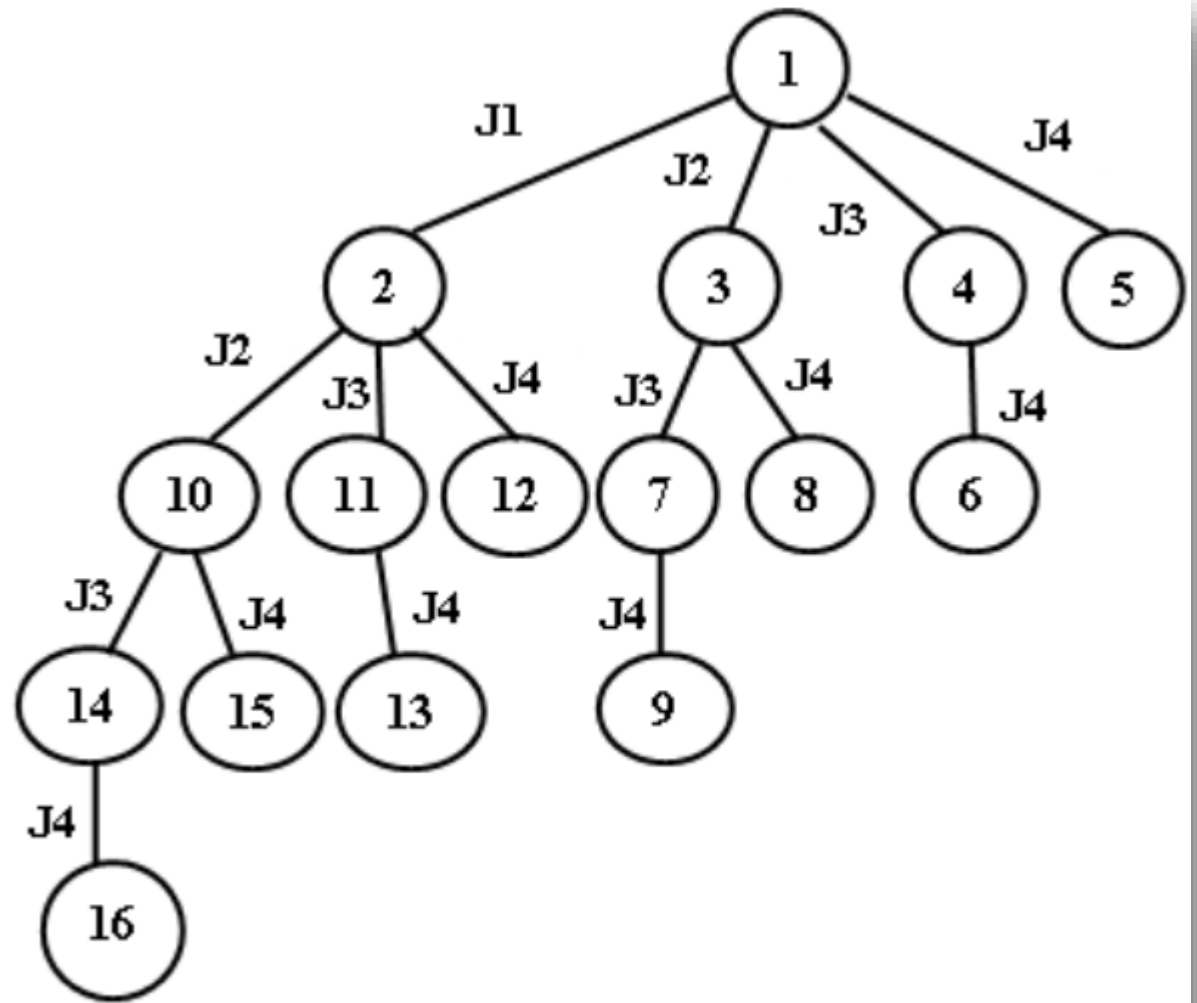
5		
4	4	6
3	3	3
2	2	2

Branch & Bound : LIFO

FIFO (First In First Out) Branch and Bound

Example: Job sequencing with deadlines problem

- Jobs = {J1, J2, J3, J4}; P = {10, 5, 8, 3}; d = {1, 2, 1, 2}
- Node 1 is the E-node. Child nodes of node 1 are generated and placed in the stack.
- First element in the stack is deleted, i.e., 5 is deleted and its child nodes are generated.
- Similarly, the next element is deleted, i.e., 4 and its child nodes are generated and placed in the queue. This is continued until an answer node is reached.



State space tree

Branch & Bound : LCBB

LC (Least Count) Branch and Bound:

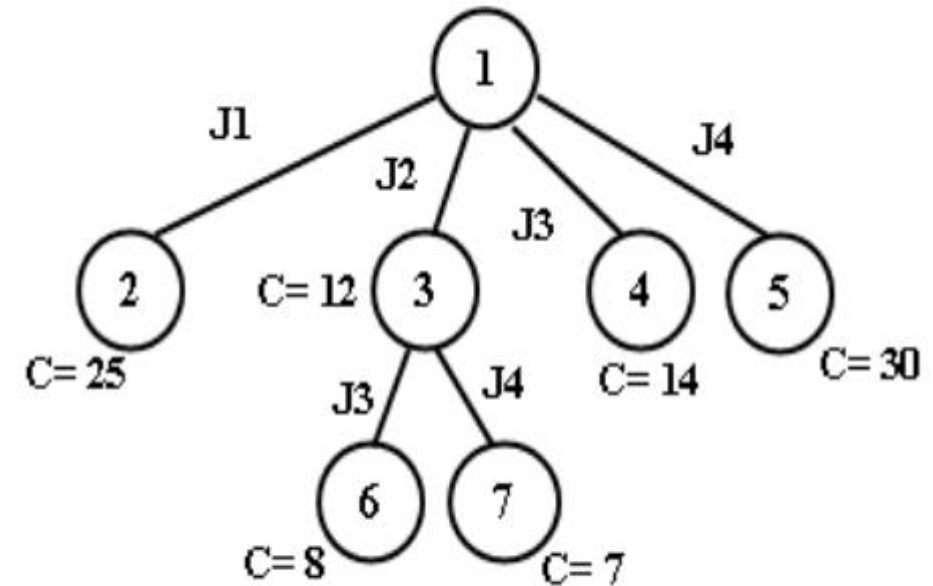
- In both FIFO and LIFO Branch and Bound the selection rules for the next E-node in rigid and blind.
- The selection rule for the next E-node does not give any preferences to a node that has a very good chance of getting the search to an answer node quickly.
- In this method ranking function or cost function is used.
- The child nodes of the E-node are generated, among these live nodes; a node which has minimum cost is selected. By using ranking function, the cost of each node is calculated.

Branch & Bound : LCBB

LC (Least Count) Branch and Bound:

Example: Job sequencing with deadlines problem

- Jobs = {J1, J2, J3, J4}; P = {10, 5, 8, 3}; d = {1, 2, 1, 2}
- Initially we will take node 1 as E-node. Generate children of node 1, the children are 2, 3, 4, 5. By using ranking function we will calculate the cost of 2, 3, 4, 5 nodes is $\hat{c} = 25$, $\hat{c} = 12$, $\hat{c} = 14$, $\hat{c} = 30$ respectively.
- Now we will select a node which has minimum cost i.e., node 3. For node 3, the children are 6, 7.



State space tree

Branch & Bound : LCBB

LC (Least Count) Branch and Bound:

- All the live nodes are stored in a PRIORITY QUEUE or HEAP.
- The live nodes are not selected according to the order in which they have been queued or stacked but according to their heuristic value.
- The heuristic value is calculated for each live node and then the node with the highest heuristic value is chosen as the E-node.

Branch & Bound : 0/1 Knapsack Problem

LC (Least Count) Branch and Bound:

- The 0/1 knapsack problem is to

$$\text{Maximize } \sum_{i=1}^n p_i x_i \quad \text{subject to } \sum_{i=1}^n w_i x_i \leq M$$

- objective of this problem is to fill the knapsack in order to maximize the profit subject to its capacity.
- But Branch & Bound is used for minimization problem.

Branch & Bound : 0/1 Knapsack Problem

LC (Least Count) Branch and Bound:

- This modified knapsack problem is stated as,
- The 0/1 knapsack problem is the maximization problem where the value of the objective function $c(x) = \sum p_i x_i$ is maximized subjected to $\sum w_i x_i \leq M$,
- Now our aim is minimization, so we take the objective function $c(x) = - \sum p_i x_i$ subjected to $\sum w_i x_i \leq M$ in order to convert the 0/1 knapsack problem as the minimization problem where $x_i = 0$ or $1, 1 \leq i \leq n$
- The two functions $\hat{c}(x)$ and $U(x)$ are defined using two algorithms Bound and UBound .

Branch & Bound : 0/1 Knapsack Problem

LC (Least Count) Branch and Bound:

- UBound computes the weights of the list of objects placed in the knapsack as a whole and their sum $\leq m$, and the profit is correspondingly decremented from initial profit and returned.
- Bound is similar to UBound but it considers fractional objects to use the entire capacity of the sack $\sum w_i x_i = m$.

Branch & Bound : 0/1 Knapsack Problem

Algorithm:

```
Algorithm Ubound(cp, cw, k, m)
{
  b = cp; c = cw;
  for i = k+1 to n do
  {
    if(c+w[i] ≤ m) then
    {
      c = c+w[i]; b = b - p[i];
    }
  }
  return b;
}
```

```
Algorithm Bound(cp, cw, k)
{
  b = cp; c = cw;
  for i = k+1 to n do
  {
    c = c+w[i];
    if(c < m) then b = p[i];
    else return b - (1 - (c - m) / w[i])*p[i];
  }
  return b;
}
```

Branch & Bound : 0/1 Knapsack Problem

$$n = 4; m = 15;$$

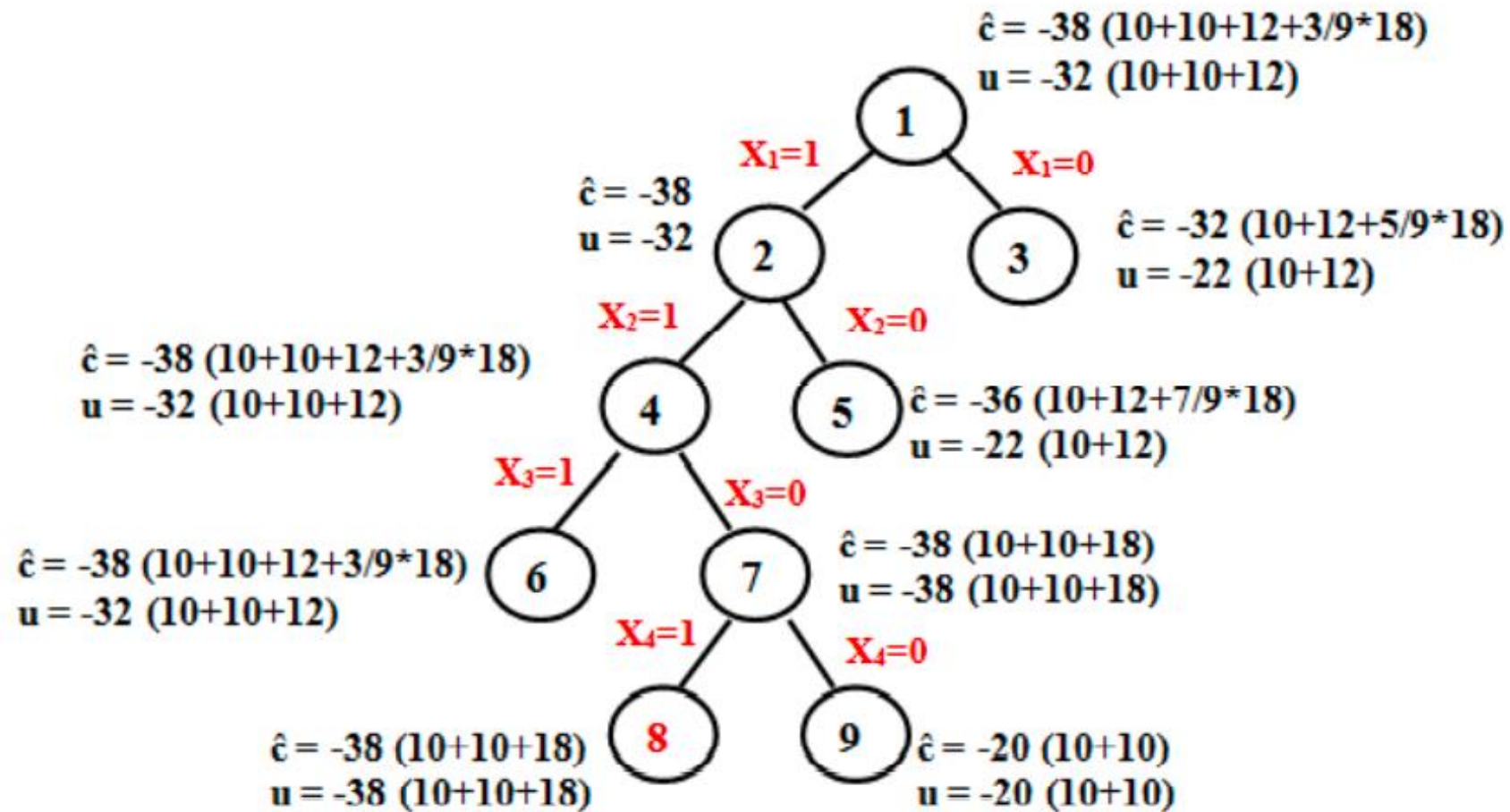
$$(p_1, p_2, p_3, p_4) = \{10, 10, 12, 18\}; (w_1, w_2, w_3, w_4) = \{2, 4, 6, 9\}$$

$$x_1 = 1,$$

$$x_2 = 1,$$

$$x_3 = 0,$$

$$x_4 = 1$$



Branch & Bound : 15 Puzzle Problem

- The 15 Puzzle problem is invented by Sam Loyd in 1878.
- The problem consist of 15 numbered (0-15) tiles on a square box with 16 tiles(one tile is blank or empty).
- The objective of this problem is to change the arrangement of initial node to goal node by using series of legal moves.
- The Initial and Goal node arrangement is shown by following figure

Initial state				Goal state			
1	2	3	4	1	2	3	4
5	6		8	5	6	7	8
9	10	7	11	9	10	11	12
13	14	15	12	13	14	15	

Figure 19 Initial and goal states for 15-puzzle problem.

Branch & Bound : 15 Puzzle Problem

- In initial node four moves are possible. User can move any one of the tile like 2, or 3, or 5, or 6 to the empty tile. From this we have four possibilities to move from initial node.
- The legal moves are for adjacent tile number is left, right, up, down, ones at a time.
- Each and every move creates a new arrangement, and this arrangement is called state of puzzle problem.
- By using different states, a state space tree diagram is created, in which edges are labeled according to the direction in which the empty space moves.
- The LCBB method is the general method used to solve the 15-puzzle problem so that the goal state can be achieved in minimum number of tile movement.

Branch & Bound : 15 Puzzle Problem

- In state space tree, nodes are numbered as per the level. In each level we must calculate the value or cost of each node by using given formula:

$$C(x)=f(x)+g(x),$$

- $f(x)$ is length of path from root or initial node to node x ,
- $g(x)$ is estimated length of path from x downward to the goal node.
Number of non-blank tile not in their correct position.
- $C(x) < \text{Infinity}$. (initially set bound).
- Each time node with smallest cost is selected for further expansion towards goal node. This node become the e-node.

Branch & Bound : 15 Puzzle Problem

- Example:

Solve the given 15-puzzle problem using LCBB.

Initial state

1	2	3	4
5	6		8
9	10	7	11
13	14	15	12

Goal state

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	

Branch & Bound : 15 Puzzle Problem

- Example:

